(ii) The orbit O is determined by the A-module structure of V, and the representation U is determined by the structure of V_x as a G_x-module.

(iii) We have

$$\sum_{U,O} \dim V^2_{(U,O)} = \sum_{U,O} |O|^2 (\dim U)^2 =$$

$$\sum_O |O|^2 |G_x| = \sum_O |O||G/G_x||G_x| = |G| \sum_O |O| = |G||A^\vee| = |G \ltimes A|.$$

(iv) The proof is essentially the same as that of the Frobenius formula. $\square$

Exercise 5.27.2. Redo Problems 4.12.1(a), 4.12.2, and 4.12.6 using Theorem 5.27.1.

Exercise 5.27.3. Deduce parts (i)—(iii) of Theorem 5.27.1 from part (iv).